Introduction to LaTeX

Chaitanya Pratap Singh

TW Lab 2025

November 11, 2026

What is LaTeX?

LaTeX is a document preparation and typesetting system commonly used for academic and technical documents.

- Produces professional-looking documents
- Useful for scientific and technical writing
- Excellent support for mathematical equations
- Automatically numbers chapters and sections
- Provides consistent document formatting

Structure and Formatting

A LaTeX document consists mainly of:

1. Document class
2. Preamble
3. Document body

Important packages used in this assignment:

- `geometry` – controls page margins
- `xcolor` – provides colors
- `sectsty` – changes heading styles
- `fancyhdr` – creates headers and footers

Conclusion

- LaTeX is useful for professional document preparation.
- It provides excellent control over document structure.
- Packages make customization simple and powerful.
- Beamer can be used to create professional presentations.

Thank You!